

Bell Nozzle Rocket Engine Calculations

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Nomenclature

C_f	Vacuum thrust coefficient
$\dot{m}$	Mass flow rate
M	Molar mass
Ma	Mach number
$mol\%$	Mole percentage
P	Pressure
R	Gas constant
R_s	Specific gas constant
T	Temperature
v	Velocity
ϵ	Nozzle area ratio
γ	Ratio of specific heats
ρ	Density

Subscripts

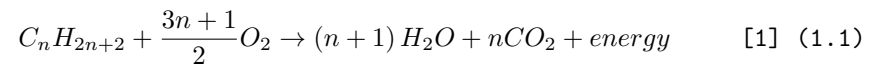
a	Ambient
c	Chamber
e	Exit
s	Stagnation
t	Throat

Introduction

Documentation of calculations for a bell nozzle rocket engine

1 Propellant

1.1 Propellant Stoichiometry of Hydrocarbon-Oxygen Fuel Mixes



1.2 Mass Flow Rate

$$\dot{m} = \frac{m}{t} \quad [2] \quad (1.2)$$

2 Exhaust Gasses

2.1 Ratio of Specific Heats of Multi-Compound Gas Mixtures

$$\frac{1}{\gamma - 1} = \sum \frac{mol\%}{\gamma_i - 1} \quad [3] \quad (2.1)$$

2.2 Stagnation Pressure of Exhaust Gasses

$$P_s = P_1 \left(1 + \left(\frac{\gamma - 1}{2} \right) Ma_1^2 \right)^{\frac{\gamma}{\gamma - 1}} \quad [4] \quad (2.2)$$

2.3 Stagnation Temperature of Exhaust Gasses

$$T_s = T_c \left(1 + \left(\frac{\gamma - 1}{2} \right) Ma_1^2 \right) \quad [5] \quad (2.3)$$

2.4 Gas Exit Velocity

$$V_e = \sqrt{\frac{2\gamma}{\gamma - 1} RT_c \left(1 - \left(\frac{P_e}{P_c} \right)^{\frac{\gamma - 1}{\gamma}} \right)} \quad [6] \quad (2.4)$$

2.5 Exhaust Gas Density

$$\rho_t = \frac{P_t}{R_s T_t} \quad [7] \quad (2.5)$$

3 Wall Geometry

3.1 Combustion Chamber Geometry

3.2 Throat Geometry

3.2.1 Nozzle Area Ratio

$$\epsilon = \frac{\left(\frac{2}{\gamma+1}\right)^{\frac{1}{\gamma-1}} \left(\frac{P_c}{P_e}\right)^{\frac{1}{\gamma}}}{\sqrt{\frac{\gamma+1}{\gamma-1} \left(1 - \left(\frac{P_e}{P_c}\right)^{\frac{\gamma-1}{\gamma}}\right)}} \quad [6] \quad (3.2.1)$$

3.2.2 Throat Area

$$A_t = \dot{m} \frac{\sqrt{T_c}}{P_c} \sqrt{\frac{R}{\gamma}} \left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \quad [8] \quad (3.2.2a)$$

$$A_t = \frac{\dot{m}}{\rho_t V_t} \quad [7] \quad (3.2.2b)$$

3.3 Nozzle Geometry

Bibliography

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